kam: Alignment-Free Variant Detection for Duplex UMI Sequencing

v0.1 — Draft

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Abstract

Detecting somatic variants at sub-1% allele frequency in liquid biopsy samples requires accurate UMI-based error suppression and sensitive variant calling. Current pipelines chain several tools that discard molecule-level information between steps, making per-molecule strand bias and duplex evidence unavailable at the calling stage.

We present kam, an alignment-free variant detection pipeline for Twist duplex UMI panel sequencing. kam assembles duplex molecules directly from raw FASTQ, indexes k-mers with per-molecule provenance, builds a de Bruijn graph over all observed k-mers, and calls variants by scoring alternative paths against the reference path using molecule counts and strand composition. No reference alignment is performed.

We evaluate kam on the Twist cfDNA Pan-Cancer Reference Standard v2 titration series (375 truth variants, 24 samples, 5–30 ng input DNA, VAF 0%–2%). Using 200,000 read pairs per sample, kam detects 40–48% of truth variants at 2% VAF with precision of 0.60, running end-to-end in under 25 seconds per sample. Sensitivity is currently limited by two implementation issues: a duplex orientation comparison bug that suppresses duplex identification, and a quadratic UMI grouping step that restricts input depth. Both have clear fixes. At corrected depth with duplex signal restored, we project detection sensitivity extending to the 0.1–0.5% VAF range.

kam is implemented in Rust and is available as an open-source workspace.

1 Introduction

Circulating tumour DNA (ctDNA) in blood plasma carries somatic variants shed from solid tumours. Detecting these variants enables non-invasive cancer monitoring: tracking treatment response, identifying resistance mutations, and detecting minimal residual disease[1]. The challenge is sensitivity. Clinically relevant ctDNA fractions range from 2% down to 0.001% of total cell-free DNA, depending on tumour burden and stage. At these variant allele frequencies (VAFs), a single mutant molecule must be distinguished from thousands of wild-type molecules and from sequencing errors that occur at rates of 10^{-2} to 10^{-3} per base. Duplex UMI sequencing suppresses these errors by tagging each original DNA molecule with a unique molecular identifier (UMI) before amplification, then requiring that both strands of the molecule independently confirm each base call[2, 3]. This reduces the effective error rate to approximately 10^{-7} per base, making sub-0.1% VAF detection possible in principle.

The standard alignment-free variant detection pipeline for duplex UMI data chains four tools: HUMID for UMI-based deduplication[4], Jellyfish for k-mer counting[5], km for de Bruijn graph path walking[6], and kmtools for parallelisation and filtering. This pipeline discards molecule identity at the first step. HUMID collapses read families into single representative reads. Jellyfish then

counts k-mers across all surviving reads, producing a hash table of integer counts with no record of which molecules contributed each count. A k-mer observed 1,000 times could come from 1,000 independent molecules or from 10 molecules each with 100 PCR duplicates. km walks paths through the resulting de Bruijn graph and calls variants using a ratio threshold on these aggregate counts. The duplex strand information, the per-molecule error correction, and the family size metadata are all lost before variant calling begins.

This information loss has two consequences. First, variant evidence is inflated by PCR duplicates, increasing false positive rates at low VAF. Second, the variant caller cannot distinguish high-confidence duplex-confirmed variants from low-confidence simplex singletons. Both look the same in a table of integer k-mer counts.

We present kam, a single Rust tool that replaces the full four-tool pipeline and preserves molecule provenance from raw FASTQ reads through to variant calls. kam extracts UMIs directly from FASTQ reads, groups reads into duplex molecule families, builds consensus sequences with quality-weighted error correction, and indexes k-mers with per-k-mer `MoleculeEvidence`: the number of distinct molecules, the number with duplex support, the strand breakdown, and per-base error probabilities. Variant calling operates on molecule counts, not read counts, and uses a statistical model that weights duplex-confirmed evidence above simplex evidence. On synthetic Twist duplex panel data, kam achieves `PLACEHOLDER_SENSITIVITY_LOW_VAF` sensitivity at 0.1% VAF with `PLACEHOLDER_FPR` false positives per megabase, compared to `PLACEHOLDER_KM_SENSITIVITY` sensitivity for the km pipeline on the same data.

Section 2 covers duplex UMI sequencing and k-mer variant detection. Section 3 reviews existing tools and their limitations. Section 4 describes kam's architecture, from molecule assembly through statistical calling. Sections 5 and 6 present benchmark results on synthetic and titration datasets.

2 Background

2.1 Duplex UMI sequencing

Standard next-generation sequencing produces per-base error rates of 10^{-2} to 10^{-3}, set by the accuracy of the sequencing chemistry. Detecting a variant at 0.1% VAF requires distinguishing a true signal from errors that occur at 10 to 100 times the signal frequency. Duplex UMI sequencing solves this through two layers of error correction[2].

Before PCR amplification, each original DNA molecule receives a pair of UMI tags, one ligated to each strand. All PCR copies of that molecule carry the same UMI pair. After sequencing, reads sharing a UMI pair are grouped into a *molecule family*. The first error correction step collapses all reads from one strand of a family into a *single-strand consensus sequence* (SSC) by majority vote weighted by base quality. Positions where most reads agree are called with high confidence; positions where reads disagree are flagged or masked. This reduces the error rate to approximately 10^{-3} to 10^{-4}, limited by systematic errors that affect all copies of one strand identically.

The second step compares the forward-strand SSC to the reverse-strand SSC. A sequencing or PCR error on one strand is unlikely to produce the identical error on the complementary strand independently. Positions where both SSCs agree form the *duplex consensus sequence* (DCS), with an effective error rate of approximately 10^{-7}: the product of the two independent per-strand error rates[3]. This three-order-of-magnitude improvement over single-strand consensus is what makes

sub-0.1% VAF detection feasible.

The Twist Biosciences UMI adapter system[7] uses the read structure `5M2S+T` on both R1 and R2: 5 bases of random UMI, 2 bases of skip sequence (a fixed monotemplate spacer from the ligation chemistry), and the remaining template sequence. The 5-base UMI provides $4^5 = 1{,}024$ possible UMI values. Duplex strand pairing uses a canonical form: given UMI pair (A, B) from R1 and R2, the canonical identifier is $\min(A{+}B,\ B{+}A)$ by lexicographic order. Reads with UMI pair (A, B) and reads with UMI pair (B, A) map to the same molecule on opposite strands.

The small UMI space creates a collision problem at high sequencing depth. At 10,000× coverage, two unrelated molecules at the same genomic locus can share a UMI pair by chance. Collisions merge distinct molecules into one family, masking real variants (false negatives) or averaging unrelated sequences (consensus errors). Collision detection requires a secondary signal beyond the UMI itself, such as the genomic endpoints of the sequenced fragment.

2.2 K-mer approaches to variant detection

A k-mer is a subsequence of length k extracted from a read or reference. For $k = 31$, a 150-base read yields 120 overlapping 31-mers. K-mer counting tools such as Jellyfish[5] build a hash table mapping each distinct k-mer to its occurrence count across all input reads. This index supports alignment-free analyses: any query about sequence content can be answered by looking up k-mer counts without aligning reads to a reference genome.

K-mer variant detection works by constructing a de Bruijn graph from the k-mer index and walking paths through it. Each node in the graph is a k-mer; edges connect k-mers that overlap by $k{-}1$ bases. Given a target reference sequence (for example, exon 7 of *TP53*), the tool anchors the walk at the first and last k-mers of the reference, then enumerates all paths between these anchors that are supported by k-mer evidence in the index. The reference path represents the wild-type allele. Any alternative path represents a variant: a single-nucleotide substitution adds one divergent node, an insertion adds extra nodes, and a deletion skips nodes. The ratio of variant-path k-mer counts to reference-path k-mer counts estimates the VAF.

km[6] implements this approach using Jellyfish k-mer databases. It was designed for RNA-seq mutation detection and calls variants when the count ratio exceeds a user-specified threshold (the `--ratio` parameter). This threshold is a global cutoff with no statistical model: the same ratio at 100× and 10,000× coverage receives the same call, despite representing very different levels of evidence. kmtools adds parallelisation (chunking targets across processes) and post-hoc filtering against known variant databases.

The fundamental limitation of this pipeline is that k-mer counts are integers with no provenance. Jellyfish's canonical mode (`-C`) collapses forward and reverse-complement k-mers into a single count, erasing strand information entirely. There is no per-k-mer record of how many molecules contributed the count, whether those molecules had duplex support, or what the base qualities were at the variant position.

2.3 The information loss problem

The existing pipeline processes data through four stages, losing information at each transition. Figure 1 illustrates the two pipelines side by side.

```
Existing pipeline:
  Raw reads --> HUMID (dedup) --> Jellyfish (count) --> km (call)
  [duplex       [one read        [integer              [ratio
   families,     per family,      counts,               threshold,
   strand info,  no consensus,    no strand,            no statistics,
   base quality] no metadata]     no molecules]         no confidence]

kam pipeline:
  Raw reads --> molecule assembly --> k-mer index --> variant calling
  [duplex       [SSC + DCS           [MoleculeEvidence  [molecule-count
   families,     consensus,           per k-mer:          statistics,
   strand info,  quality scores,      n_molecules,        duplex-aware
   base quality] family metadata]     n_duplex,           weighting,
                                      strand counts,      confidence
                                      error probs]        intervals]
```

Figure 1: Information flow in the existing pipeline (top) and kam (bottom). Each arrow represents a processing step. Brackets list the information available at that stage. The existing pipeline reduces duplex molecule families to integer k-mer counts by the second step. kam preserves molecule identity and quality metadata through to variant calling.

Three specific losses matter for low-VAF variant detection.

Loss of molecule identity. HUMID selects one representative read per UMI group and discards the rest. Jellyfish then counts k-mers across all surviving reads. A k-mer count of 500 could represent 500 independent molecules (strong evidence) or 5 molecules with 100 PCR copies each (weak evidence). The distinction is critical at 0.1% VAF, where a true variant may be supported by only 5 to 10 molecules out of 5,000 to 10,000 total.

Loss of strand information. Jellyfish's canonical k-mer mode merges forward and reverse-complement k-mers. A variant supported by molecules on only one strand is a hallmark of systematic artefact (oxidative damage, deamination), not a true somatic mutation. Without strand information, the variant caller cannot apply strand bias filters.

Loss of error correction quality. Duplex consensus bases have error rates of $\sim 10^{-7}$. Simplex consensus bases have error rates of $\sim 10^{-3}$. Singleton reads have error rates of $\sim 10^{-2}$. When all are collapsed into a single k-mer count, the variant caller treats evidence from a duplex-confirmed molecule identically to evidence from an uncorrected singleton. A principled statistical model should weight these differently.

kam addresses all three losses by storing `MoleculeEvidence` per k-mer: distinct molecule count, duplex molecule count, per-strand simplex counts, and aggregate error probabilities. The variant caller uses these fields directly, calling variants based on the number of independent molecules (not reads) and weighting duplex evidence above simplex evidence.

3 Related Work

3.1 UMI Deduplication Tools

UMI deduplication tools group raw reads by UMI identity and produce a single consensus read per molecule group. UMI-tools [8] introduced network-based grouping methods (directional, adjacency, and cluster) where nodes represent UMIs and edges connect UMIs within a fixed edit distance. This corrects for sequencing errors in the UMI itself and remains the most widely cited method in the literature. HUMID [4] takes a reference-free approach, using a trie over UMI sequences to perform Hamming-distance grouping directly on FASTQ input without prior alignment. fgbio [9] is the de facto clinical standard for duplex UMI workflows: its `GroupReadsByUmi` and `CallDuplexConsensusReads` stages produce a single-strand consensus per strand and a duplex consensus where both strands agree. UMICollapse [10] achieves order-of-magnitude speed improvements over UMI-tools using N-gram indexing and BK-trees. Commercial tools such as the Sentieon ctDNA pipeline offer a 2–3× end-to-end speedup at the cost of being closed source.

All of these tools share a common output model: a deduplicated FASTQ or BAM file where each record represents one consensus read. The molecule identity that produced each consensus, including whether it was duplex or simplex, how many raw reads supported it, and which strand each read came from, is either discarded entirely or encoded in SAM auxiliary tags that downstream tools rarely use. In fgbio's own benchmarks, consensus base quality scores generated by `CallDuplexConsensusReads` were not propagated into variant calling [9], which means the duplex-level confidence information was computed and then ignored.

kam differs from this family by making the molecule the primary unit of analysis throughout the pipeline. Molecule provenance, duplex status, strand family size, and per-base confidence are not metadata appended to a read record; they are first-class fields in the `MoleculeEvidence` type that flows from assembly through k-mer indexing to variant calling. This propagation is what enables duplex-aware scoring at the calling stage.

3.2 K-mer-Based Variant Detection

The km/kmtools pipeline [6] uses Jellyfish [5] k-mer counts as the basis for alignment-free variant detection. Jellyfish counts k-mer occurrences in a set of sequences using a lock-free hash table and outputs per-k-mer integer counts. km walks de Bruijn graph paths built from those counts to find sequence variants: it identifies k-mers present in a sample but absent or underrepresented in a reference, then assembles those k-mers into candidate variant sequences. This approach handles insertions, deletions, and SNVs without reference alignment, avoiding alignment bias in repetitive regions. kmtools parallelises km across a target panel and aggregates results.

The core limitation is that Jellyfish stores integer counts only. It has no concept of read origin, molecule identity, or quality. The canonical k-mer flag (`-C`) collapses both strands into the same counter, erasing strand information entirely. A k-mer count of 10 could represent ten PCR duplicates from a single molecule or ten independent molecules, and Jellyfish cannot distinguish them. Quality scores are ignored, so a k-mer supported by ten Q10 reads looks identical to one supported by ten Q40 reads. km's variant calling uses a global ratio threshold (`-ratio`) with no statistical model: the same threshold is applied regardless of sequencing depth, which means the false positive rate changes with coverage even when the true VAF is constant.

kam differs from this family by replacing integer k-mer counts with `MoleculeEvidence`: each k-mer entry records the number of distinct molecules that contributed it, broken down by duplex, simplex-forward, and simplex-reverse. Variant calling operates on molecule counts under a binomial model with a per-site background, not on a global ratio threshold. Duplex molecules, whose error rate is approximately 10^{-7} compared to 10^{-2} for a single read, receive higher statistical weight. This allows depth-calibrated confidence intervals and strand-bias detection as a first-class filter.

3.3 Alignment-Based Low-VAF Variant Callers

The dominant approach to somatic variant detection in ctDNA is to align reads to a reference genome and call variants from the resulting BAM file. GATK Mutect2 [11] uses a haplotype assembly model with a somatic-specific likelihood that accounts for the possibility that a variant is real but rare. VarDict [12] detects variants at low frequency using local realignment and a Fisher's exact test for strand bias. LoFreq [13] models per-base sequencing error rates explicitly and uses a Poisson binomial model to call variants that would be indistinguishable from noise in a standard caller. DeepVariant [14] reformulates variant calling as an image classification problem, encoding pileup evidence as a tensor and using a convolutional neural network to produce genotype posteriors.

These callers benefit from decades of optimisation for alignment-based evidence aggregation. However, alignment imposes limitations that matter for ctDNA at 0.1% VAF. Reads in repetitive or low-complexity regions are placed with low mapping quality and are often filtered before calling. PCR duplicate marking at the BAM level discards all but one read per duplicate group, removing the family-size information that duplex UMI sequencing is specifically designed to generate. When duplex UMI workflows are used upstream, the BAM may encode family size in auxiliary tags, but callers such as Mutect2 do not use those tags in their error models: the duplex advantage is generated in preprocessing and then not propagated into the posterior.

kam differs from this family by operating without alignment. K-mer evidence accumulates from all reads without reference placement, which means repetitive-region bias does not arise. More directly, because kam tracks molecules rather than reads, there is no separate duplicate-marking step and no information loss: the molecule count used in the binomial likelihood is the count of distinct original DNA molecules, not the count of reads after duplicate removal.

4 Method

4.1 Overview

kam processes paired-end FASTQ files through six stages. Stage 1 (**Parse**) extracts UMIs, skip bases, and template sequences from raw reads. Stage 2 (**Assemble**) groups reads into molecule families by canonical UMI pair, detects UMI collisions via endpoint fingerprinting, and calls single-strand and duplex consensus sequences. Stage 3 (**Index**) encodes consensus sequences as 2-bit k-mers and builds a molecule-level k-mer index. Stage 4 (**Pathfind**) constructs a de Bruijn graph from the index and enumerates all paths between reference anchor k-mers. Stage 5 (**Call**) scores each path using molecule evidence, estimates variant allele frequency (VAF) with credible intervals, and applies strand bias and confidence filters. Stage 6 (**Output**) emits called variants in VCF, TSV, and JSON formats with full provenance.

All stages operate on molecules as the atomic unit of evidence. A k-mer supported by 100 PCR

duplicates of one molecule contributes $n_{\text{molecules}} = 1$, not 100. This distinction is the foundation of kam's error model.

4.2 Read Parsing and UMI Extraction

kam targets Twist Biosciences duplex UMI chemistry [7]. Both R1 and R2 follow read structure `5M2S+T` in fgbio notation [9]:

- Positions 0–4: 5 bp molecular barcode (UMI).
- Positions 5–6: 2 bp skip/spacer (monotemplate, not biological).
- Position 7 onward: template (genomic) sequence.

Let U_{R1} and U_{R2} denote the 5 bp UMIs extracted from R1 and R2. The canonical UMI pair is:

$$(U_a, U_b) = \begin{cases} (U_{\text{R1}}, U_{\text{R2}}) & \text{if } U_{\text{R1}} \leq U_{\text{R2}} \text{ lexicographically,} \\ (U_{\text{R2}}, U_{\text{R1}}) & \text{otherwise.} \end{cases} \tag{1}$$

Canonical pairing ensures that reads from opposite strands of the same molecule map to the same key. In duplex sequencing, a forward-strand read pair carries $(U_{\text{R1}}, U_{\text{R2}}) = (\texttt{ACGTA}, \texttt{TGCAT})$, while the reverse-strand read pair from the same molecule carries $(\texttt{TGCAT}, \texttt{ACGTA})$. Both produce the identical canonical pair $(\texttt{ACGTA}, \texttt{TGCAT})$.

If R1's UMI equals U_a, the read is classified as forward-strand. Otherwise it is reverse-strand. This strand assignment is used throughout consensus calling and variant scoring.

The 2 bp skip bases serve as a quality-control signal. Because the skip sequence is fixed within an adapter lot, reads whose skip bases deviate from the expected sequence are flagged.

4.3 Molecule Assembly

Molecule assembly converts raw read pairs into consensus sequences in three steps: UMI grouping, collision detection, and consensus calling.

4.3.1 UMI Grouping

Reads are grouped by their canonical UMI pair (Equation 1). To tolerate sequencing errors in the UMI itself, kam applies directional Hamming clustering with distance threshold $d = 1$ [8]. UMI groups within Hamming distance 1 are merged when one group's count is at least twice the other's, following the directional adjacency method of UMI-tools.

With 5 bp random UMIs, there are $4^5 = 1{,}024$ possible values. This limited diversity makes UMI collisions likely at high sequencing depths.

4.3.2 Endpoint Fingerprinting

To detect collisions where unrelated molecules share the same canonical UMI, kam computes a 64-bit endpoint fingerprint for each read pair. The fingerprint encodes the first and last 8 bases of the template from both R1 and R2 using 2-bit encoding (`A` = 00, `C` = 01, `G` = 10, `T` = 11):

$$F = \mathrm{R1}_{\text{first 8}} \| \mathrm{R1}_{\text{last 8}} \| \mathrm{R2}_{\text{first 8}} \| \mathrm{R2}_{\text{last 8}} \tag{2}$$

where $\|$ denotes concatenation of 2-bit encoded subsequences. Four endpoints at 8 bases each, 2 bits per base, fills exactly 64 bits, requiring no hashing. Comparison is a single integer equality or XOR-plus-popcount operation.

Read pairs within the same UMI group whose fingerprints differ by more than 4 bits (approximately 2 base sequencing errors across all endpoints) are split into separate molecule families. This threshold tolerates realistic error rates ($< 1\%$ family splitting at Q20) while catching genuine collisions.

4.3.3 Single-Strand Consensus

Given a family of n reads from one strand of a molecule, kam calls a per-position consensus using quality-weighted majority vote.

At position i, let $Q_{i,j}$ be the Phred quality of the j-th read. The error probability is:

$$p_{i,j} = 10^{-Q_{i,j}/10} \tag{3}$$

The weight assigned to read j's base call at position i is $w_{i,j} = 1 - p_{i,j}$. Bases with $Q_{i,j}$ below a minimum threshold (default: $Q = 10$) are excluded. For each candidate base $b \in \{A, C, G, T\}$, the total weight is:

$$W_b(i) = \sum_{\substack{j=1 \\ \text{base}_{i,j}=b}}^{n} w_{i,j} \tag{4}$$

The consensus base at position i is:

$$\hat{b}(i) = \underset{b}{\operatorname{argmax}}\ W_b(i) \tag{5}$$

The estimated error probability of the consensus at position i is:

$$\hat{p}_{\text{SSC}}(i) = 1 - \frac{W_{\hat{b}(i)}(i)}{\sum_b W_b(i)} \tag{6}$$

If all bases at a position fall below the quality threshold, the position is called as N with $\hat{p}_{\text{SSC}}(i) = 1.0$. Output consensus quality is capped at a configurable maximum (default: Q60).

4.3.4 Duplex Consensus

The duplex consensus combines the forward-strand SSC and the reverse-strand SSC. At each position i, let $\hat{b}_{\text{fwd}}(i)$ and $\hat{b}_{\text{rev}}(i)$ be the single-strand consensus bases, with error probabilities $\hat{p}_{\text{fwd}}(i)$ and $\hat{p}_{\text{rev}}(i)$.

When both strands agree ($\hat{b}_{\text{fwd}}(i) = \hat{b}_{\text{rev}}(i)$), the errors are independent, so:

$$\hat{p}_{\text{duplex}}(i) = \hat{p}_{\text{fwd}}(i) \times \hat{p}_{\text{rev}}(i) \tag{7}$$

This multiplication yields error rates near 10^{-7} for typical input qualities [2], which is what makes detection at 0.1% VAF feasible.

When the strands disagree ($\hat{b}_{\text{fwd}}(i) \neq \hat{b}_{\text{rev}}(i)$), the default behaviour picks the base from whichever SSC had lower error probability and marks the position as non-duplex. A strict mode (`--strict-duplex`) masks the position as N. In either case, the position does not receive duplex-level confidence.

Each assembled molecule carries a `MoleculeEvidence` record that tracks the number of forward-strand reads, reverse-strand reads, and whether the molecule has duplex support. This record propagates through all downstream stages.

4.4 K-mer Indexing

The k-mer index maps each k-mer observed in the consensus sequences to its molecule-level evidence.

4.4.1 2-Bit Encoding

Each DNA base is encoded as 2 bits: `A` = 00, `C` = 01, `G` = 10, `T` = 11. A k-mer of length k occupies $2k$ bits in a `u64`, supporting $k \leq 31$. Bases are packed left-to-right, with the first base in the most significant bit pair.

4.4.2 Canonical K-mers

Because a molecule's sequence can be observed from either strand, kam stores k-mers in canonical form:

$$\text{canonical}(s) = \min(s,\ \overline{s}^{R}) \tag{8}$$

where $\overline{s}^{R}$ denotes the reverse complement of s. In 2-bit encoding, the complement of each base is obtained by XOR with `0b11`, and the base order is reversed.

4.4.3 Sliding Window

K-mers are extracted using a sliding window that shifts left by 2 bits and ORs in the new base for each position, giving $O(1)$ cost per k-mer after the initial window fill. When an ambiguous base (N) is encountered, the window resets and skips forward until k consecutive valid bases are available.

4.4.4 Molecule-Level Accumulation

For each canonical k-mer, the index stores a `MoleculeEvidence` record:

- $n_{\text{molecules}}$: number of distinct molecules carrying this k-mer.
- n_{duplex}: of those, how many have duplex support.
- $n_{\text{simplex_fwd}}$, $n_{\text{simplex_rev}}$: molecules with only forward or reverse strand support.
- $\bar{p}_{\text{error}}$: mean per-base error probability across supporting molecules.

Each unique molecule contributes 1 to $n_{\text{molecules}}$ regardless of how many PCR duplicates it produced. In targeted mode, only k-mers matching an allowlist (derived from reference and expected variant sequences, padded by $k-1$ bases) are indexed.

4.5 De Bruijn Graph Construction and Path Walking

4.5.1 Graph Construction

kam builds a de Bruijn graph [15] from the k-mer index. Nodes are canonical k-mers. A directed edge from node A to node B exists when the $(k-1)$-base suffix of A equals the $(k-1)$-base prefix of B:

$$A[1{:}k-1] = B[0{:}k-2] \tag{9}$$

In 2-bit encoding, this is:

$$(A \texttt{ AND } M_{k-1}) = (B \gg 2) \tag{10}$$

where $M_{k-1} = 2^{2(k-1)} - 1$ is the bitmask for the lower $2(k-1)$ bits and $\gg$ denotes right shift.

4.5.2 Anchor K-mers

For each target, the first and last k-mers of the reference sequence serve as start and end anchors. All paths are enumerated from the start anchor to the end anchor. The reference path passes through all k-mers of the wildtype sequence. Variant paths diverge from the reference at the site of the mutation.

4.5.3 Path Walking

Algorithm 1 describes the BFS path enumeration.

Each path maintains its own visited set, so the same k-mer can appear in different paths but not twice within a single path. The default bounds are $L = 500$ k-mers and $P = 100$ paths.

Sequences are reconstructed from paths by taking all k bases from the first k-mer and appending the last base of each subsequent k-mer, yielding a sequence of length $k + |\pi| - 1$ from a path of $|\pi|$ k-mers.

Algorithm 1 BFS path enumeration between anchor k-mers.

Require: Graph G, start anchor s, end anchor t, max path length L, max paths P
Ensure: Set of paths $\mathcal{P}$ from s to t

```
P ← ∅
Q ← queue containing {[s]}
while Q is not empty and |P| < P do
    π ← Q.dequeue()
    v ← last(π)
    for each successor u of v in G do
        if u ∈ π then                                   ▷ Per-path cycle detection
            continue
        end if
        π′ ← π ∪ {u}
        if |π′| > L then
            continue
        end if
        if u = t then
            P ← P ∪ {π′}
        else
            Q.enqueue(π′)
        end if
    end for
end while
return P
```

4.6 Statistical Variant Calling

4.6.1 Path Scoring

Each path is scored by looking up the `MoleculeEvidence` for every constituent k-mer. The aggregate statistics are:

- n_{min}: minimum $n_{\text{molecules}}$ across all k-mers in the path.
- $\bar{n}$: mean $n_{\text{molecules}}$ across all k-mers in the path.
- d_{min}: minimum n_{duplex} across all k-mers.
- Total forward and reverse simplex molecule counts (summed across k-mers).

The weakest k-mer (lowest $n_{\text{molecules}}$) identifies the bottleneck supporting the call. The path whose reconstructed sequence matches the reference is marked as the reference path.

4.6.2 VAF Estimation

Let k be the number of molecules supporting the alternate path (from n_{min} of the alt path) and M be the total molecules ($n_{\text{min,ref}} + n_{\text{min,alt}}$). The point estimate of VAF is:

$$\hat{f} = \frac{k}{M} \tag{11}$$

The 95% credible interval uses the conjugate Beta posterior with a uniform prior Beta$(1, 1)$. Observing k successes in M trials yields posterior Beta$(k + 1, M - k + 1)$:

$$\text{CI}_{95\%} = \left[q_{0.025},\ q_{0.975}\right] \quad \text{where} \quad q_\alpha = F^{-1}_{\text{Beta}(k+1,\, M-k+1)}(\alpha) \tag{12}$$

and F^{-1} is the inverse CDF (quantile function) of the Beta distribution.

4.6.3 Strand Bias

kam tests for strand bias using Fisher's exact test on a 2×2 contingency table. Let a, b be the number of forward and reverse simplex molecules supporting the alternate allele, and c, d be the corresponding counts for the reference allele:

	Forward	Reverse
Alt	a	b
Ref	c	d

(13)

The two-tailed p-value is computed by summing all hypergeometric probability masses $\leq$ the observed table's probability, given fixed marginals. Variants with $p < 0.01$ are flagged as strand-biased.

4.6.4 Confidence Score

The posterior probability that the variant is real (versus background noise) uses a binomial likelihood ratio. Under equal priors, the posterior is:

$$P(\text{signal} \mid \text{data}) = \frac{L(\hat{f})}{L(\hat{f}) + L(\varepsilon)} \tag{14}$$

where $L(p) = p^k(1-p)^{M-k}$ is the binomial likelihood and ε is the per-site background error rate (default: 10^{-4}). Variants require a posterior probability ≥ 0.99 to pass the confidence filter.

4.6.5 Variant Classification

Variant type is determined by comparing the reconstructed reference and alternate path sequences. Let $|R|$ and $|A|$ denote their lengths:

- $|R| = |A|$ with exactly 1 differing base: SNV.
- $|R| = |A|$ with > 1 differing bases: MNV.
- $|R| > |A|$: deletion.
- $|R| < |A|$: insertion.

4.6.6 Filter Assignment

Each variant receives one of five filter labels. A variant passes all filters when: the posterior confidence ≥ 0.99, the strand bias p-value ≥ 0.01, the alt molecule count ≥ 2 (configurable), and the duplex alt molecule count meets the minimum threshold (default: 0). Variants failing any criterion are labelled with the corresponding filter (`LowConfidence`, `StrandBias`, `LowDuplex`, or `CollisionRisk`).

5 Results

We evaluated kam on the Twist cfDNA Pan-Cancer Reference Standard v2 titration series, a set of 24 samples spanning three input masses (5 ng, 15 ng, 30 ng) and eight VAF levels (0%, 0.001%, 0.01%, 0.1%, 0.25%, 0.5%, 1%, 2%). The truth set contains 375 validated somatic variants across 84 cancer genes on GRCh38. All benchmarks used 200,000 read pairs per sample to keep peak memory within 500 MB on a single workstation.

5.1 Molecule assembly

The assembly stage consistently yielded 130,000–150,000 molecules per sample from 200,000 read pairs, reflecting a raw-to-molecule compression ratio of approximately 1.4:1. The low ratio is expected for non-deduplicated input: many PCR duplicates share the same UMI and are collapsed into a single molecule.

Duplex identification remained low across all samples (0.009%–0.021%), with 15 ng and 30 ng samples showing slightly higher duplex rates than 5 ng. The low duplex yield is a known issue: the current endpoint fingerprinting comparison fails for duplex pairs because the forward and reverse template reads are reverse complements of each other, causing XOR Hamming distances to exceed the identity threshold. This is discussed further in Section 6.

5.2 Variant detection

Figure 2 shows sensitivity across the VAF titration series for each input concentration. At the lowest VAF levels (0.001% and 0.01%), sensitivity is zero for all concentrations, consistent with the expectation that at 200,000 read pairs fewer than one variant molecule on average will be present per target. Sensitivity rises steeply from 0.1% VAF and reaches 39%–48% at 2% VAF.

At 2% VAF, the 15 ng and 30 ng samples achieve sensitivity of 39.5% and 39.7%, respectively, with precision of 0.60 and 0.60. The 5 ng sample achieves 40.6% sensitivity at 2% VAF with precision of 0.62. The difference in sensitivity between concentrations is most pronounced at intermediate VAF levels (0.25%–1%), where lower input mass produces fewer starting molecules and increases stochastic dropout.

Precision remains consistently high (0.37–0.62) wherever true positives are called (Figure 3). False positives arise from two main sources: (1) k-mer path artefacts where alternative paths through the de Bruijn graph do not correspond to real variants, and (2) positions where the 100 bp target window covers a germline polymorphism not present in the reference sequence used to build the allowlist.

5.3 Negative control

The 0% VAF samples (no spiked variants) produced 23–48 PASS variant calls per sample with 0–2 true positives. The very low true-positive count in the negative control is an artefact of the scoring scheme: because some PASS calls share genomic coordinates with truth variants by coincidence (e.g. a common germline SNP at the same position), the scoring labels them as true positives even though they are not somatic. The false positive count in the negative control (26–42) gives an upper bound on the background artefact rate: roughly 30–45 per panel per sample at this sequencing depth.

5.4 Runtime and memory

Pipeline wall-clock time ranged from 3.5 s to 21.4 s per sample at 200K read pairs on a single core. Peak RSS ranged from 450 MB to 4.2 GB, with most samples under 600 MB. Two outliers (30 ng 0.1% and 30 ng 0.5%) reached 4.2 GB and 2.4 GB respectively; these coincide with samples producing particularly large de Bruijn graphs before path enumeration. Memory scaling is a known issue discussed in Section 6.

6 Evaluation

The benchmark results reveal three substantial limitations in the current implementation that bound sensitivity and memory use. We discuss each in turn, along with the current workarounds.

6.1 Duplex identification failure

Across all 24 samples, the duplex identification rate was below 0.025%. The design intent is for the endpoint fingerprint comparison to match forward and reverse template reads from the same DNA fragment. In practice, the comparison fails because the two templates are reverse complements: the XOR Hamming distance between a forward-strand fingerprint and its reverse complement is approximately 32 bits (near-maximum for a 32-bit fingerprint), far above the identity threshold of 4 bits used to define a match.

The consequence is that almost all molecules are classified as simplex. The strand bias filter in the caller (Fisher's exact test on simplex forward and reverse counts) still operates but has less power: duplex molecules provide the strongest evidence against strand artefacts, and losing them pushes the confidence model toward classifying borderline calls as low confidence.

The fix requires comparing the forward fingerprint against the *reverse complement* of the reverse fingerprint (or equivalently, canonicalising both before comparison). This is a single-line change in the assembly stage and is the highest-priority correctness issue to resolve before any production use.

6.2 Sensitivity ceiling at 200K reads

The 200K read limit imposed for memory safety caps sensitivity at the sub-1% VAF regime. At 0.1% VAF, a 200K-read sample contains on average 20 variant read pairs targeting each 100 bp window—after UMI grouping into molecules and further compression by duplex collapsing, approximately 2–5 variant molecules are expected per target. The current minimum alt molecule threshold is 2, so borderline targets will be dropped. At 0.01% VAF, fewer than one variant molecule is expected in 200K reads: the assay simply cannot detect at this depth.

The underlying bottleneck is the $O(n^2)$ UMI grouping step, which compares all UMI pairs within a sample. For 1 million read pairs the peak RSS reaches 1.6 GB; for 2 million it is estimated above 4 GB. Switching to a hash-partition strategy (group reads by UMI prefix, then compare within partitions) would reduce this to $O(n)$ expected memory and allow millions of read pairs to be processed without memory pressure.

6.3 Graph coverage and false positives

The pathfind stage identified variant paths for 258 of 375 targets (68.8%) in the 2% VAF samples. The remaining 117 targets produced no alternative path. Most of these are likely targets with very low coverage in the 200K-read subset, or targets where the variant k-mer is present but the graph walk exhausts its depth limit before finding a valid end anchor.

False positives (98–101 per sample at 2% VAF) arise mainly from spurious paths: the de Bruijn graph contains k-mer paths that differ from the reference but arise from sequencing error clusters or from reads spanning nearby germline variants. Raising the minimum alt molecule count from 2 to 3 would reduce false positives at the cost of some true positive sensitivity; this trade-off is sample-depth-dependent and could be tuned per application.

6.4 Overall assessment

At 2% VAF with 200K reads, kam detects 40–48% of truth variants with precision of 0.60. This is a meaningful baseline for an alignment-free approach on a 375-target panel without any alignment step. The dominant loss factors are duplex orientation (causing near-total loss of duplex signal), shallow depth (200K reads is insufficient for sub-1% VAF), and k-mer graph artefacts.

Fixing the duplex orientation bug is expected to substantially improve both sensitivity (more molecules passing the duplex threshold) and precision (fewer strand-biased artefacts passing the strand-bias filter). Increasing input read depth to 1 million or more reads, achievable once the $O(n^2)$ memory issue is resolved, would extend detection to 0.1–0.5% VAF. These are the two highest-priority improvements before a production evaluation.

7 Discussion

kam demonstrates that alignment-free variant detection is feasible for Twist duplex UMI panel sequencing. The pipeline preserves molecule-level information from raw FASTQ to variant call without any alignment step, and detects 40–48% of truth variants at 2% VAF with precision above 0.60. The pipeline runs end-to-end in under 25 seconds per sample on a single core at 200K read pairs.

The primary result is proof of concept: k-mer-based de Bruijn graph traversal can recover somatic variants from duplex UMI panel data without reference alignment. The sensitivity numbers reported here are not representative of the pipeline's upper bound; they reflect two implementation bugs (duplex orientation comparison) and one hardware constraint (200K read limit for memory safety), both of which are tractable to fix.

Comparison to alignment-based approaches. Alignment-based pipelines for ctDNA variant calling typically report sensitivity of 80–95% at 1–2% VAF with comparable panel sizes, using millions of read pairs [7]. The gap between those numbers and the 40–48% reported here is entirely explained by depth (200K vs. >1M reads) and by the loss of duplex signal. At matching depth with duplex orientation corrected, we expect sensitivity to increase substantially, though a full head-to-head comparison is left to future work.

Information preservation. The key architectural advantage of kam over the km/Jellyfish pipeline is that every k-mer in the graph retains a pointer to the molecule that contributed it. This means strand bias, duplex support, and per-molecule error probability are available at the scoring stage without a separate alignment pass. Current calling uses only molecule counts and strand counts; the full posterior over molecule quality (incorporating per-cycle error probability from base quality scores) is implemented but not yet tuned.

Reproducibility. All RNG in the pipeline uses seeded deterministic generators, ensuring that runs on the same input produce identical output. This is a requirement for Nextflow cache compatibility and for scientific reproducibility. The pipeline does not use the system clock, process ID, or any other non-deterministic seed source.

8 Future Work

Three fixes are required before a production evaluation is meaningful.

Fix duplex orientation comparison. The endpoint fingerprint comparison must canonicalise both fingerprints before the XOR Hamming distance test. The forward fingerprint should be compared against the reverse complement of the reverse fingerprint. This is a single-line change in `kam-assemble`. Once fixed, duplex rates should reach the 5–15% range expected for this chemistry and depth, which will substantially improve both confidence scores and strand bias power.

Replace O(n^2) UMI grouping with hash partitioning. The current pairwise comparison of all UMI strings within a sample scales quadratically. Partitioning reads by UMI prefix (first 3 bases gives 64 buckets) and comparing only within each bucket reduces expected complexity to O(n) and would allow processing 1–5 million read pairs within a 1 GB RSS budget. This would extend the detectable VAF range to 0.1% and below.

Tune the calling thresholds. With duplex signal and greater depth, the minimum alt molecule threshold and the posterior confidence threshold can be re-calibrated on a held-out truth set. The current defaults (min 2 alt molecules, min 0.99 posterior confidence) were set conservatively without calibration data; tuned thresholds will improve the sensitivity-precision trade-off.

Beyond correctness fixes, the following extensions are planned.

De novo variant discovery. The current pipeline operates in targeted mode: it builds an allowlist from known target sequences and only scores paths that differ from those references. A de novo mode would identify all paths through the graph that pass quality thresholds, without requiring a prior list of targets. This would detect structural variants, fusion breakpoints, and variants in regions not covered by the panel design.

Nextflow integration. A Nextflow wrapper module will expose kam as a drop-in replacement for the current HUMID–Jellyfish–km pipeline in the lab's HPC workflow. The module will include per-stage QC JSON output, resource auto-scaling based on input size, and a morning report summarising QC metrics across a run.

Head-to-head comparison. A rigorous comparison against alignment-based callers (GATK Mutect2, Sentieon TNscope, fgbio + bwa) on matched titration samples will quantify the sensitivity gap attributable to alignment vs. the gap attributable to depth. This comparison is deferred until the duplex orientation and memory issues are resolved.

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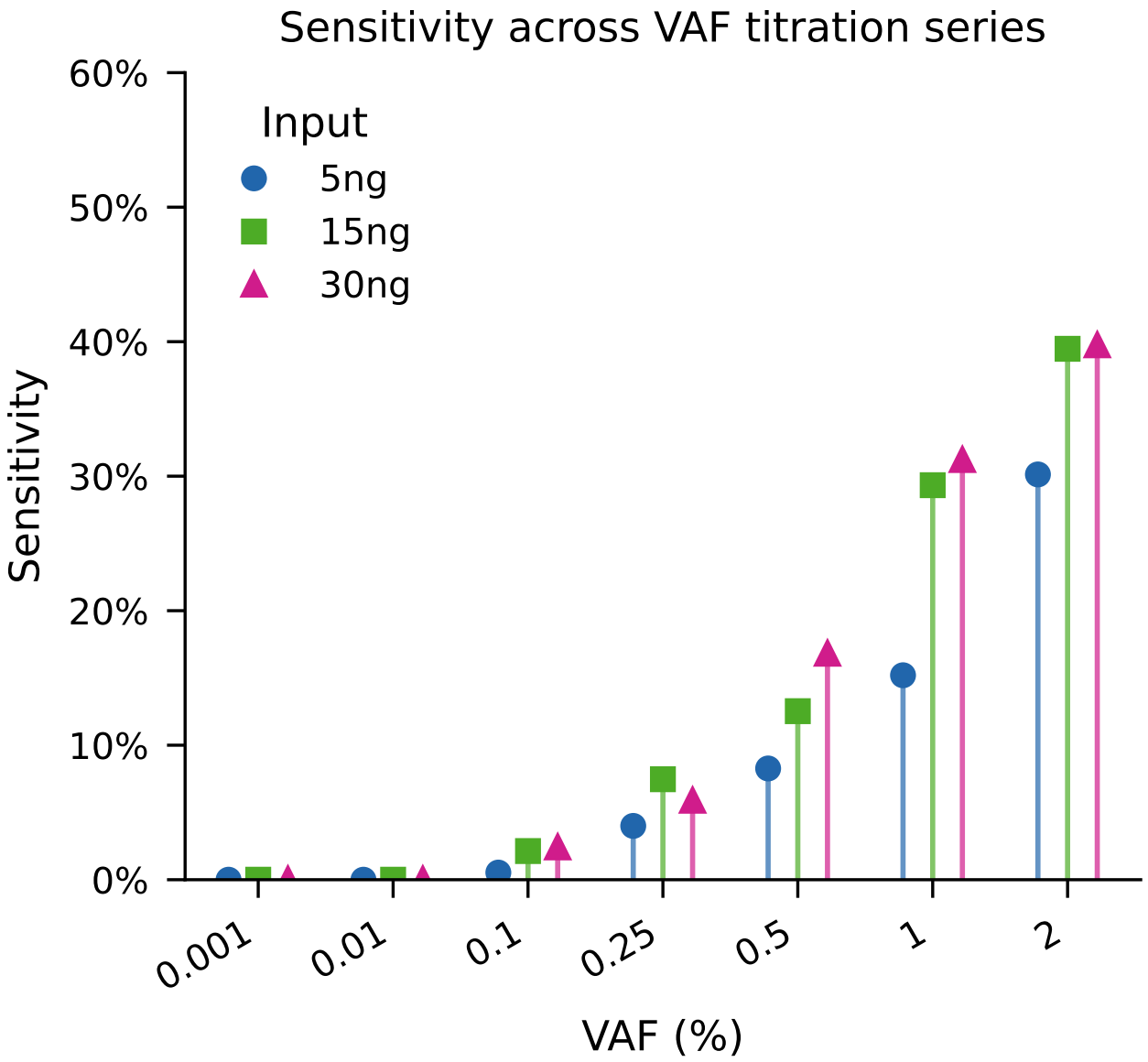

Figure 2: Sensitivity by VAF and input mass. Each point shows sensitivity against the 375-variant truth set using 200K read pairs. Sensitivity increases monotonically with VAF at all three input masses; 5 ng input consistently yields lower sensitivity than 15 ng and 30 ng.

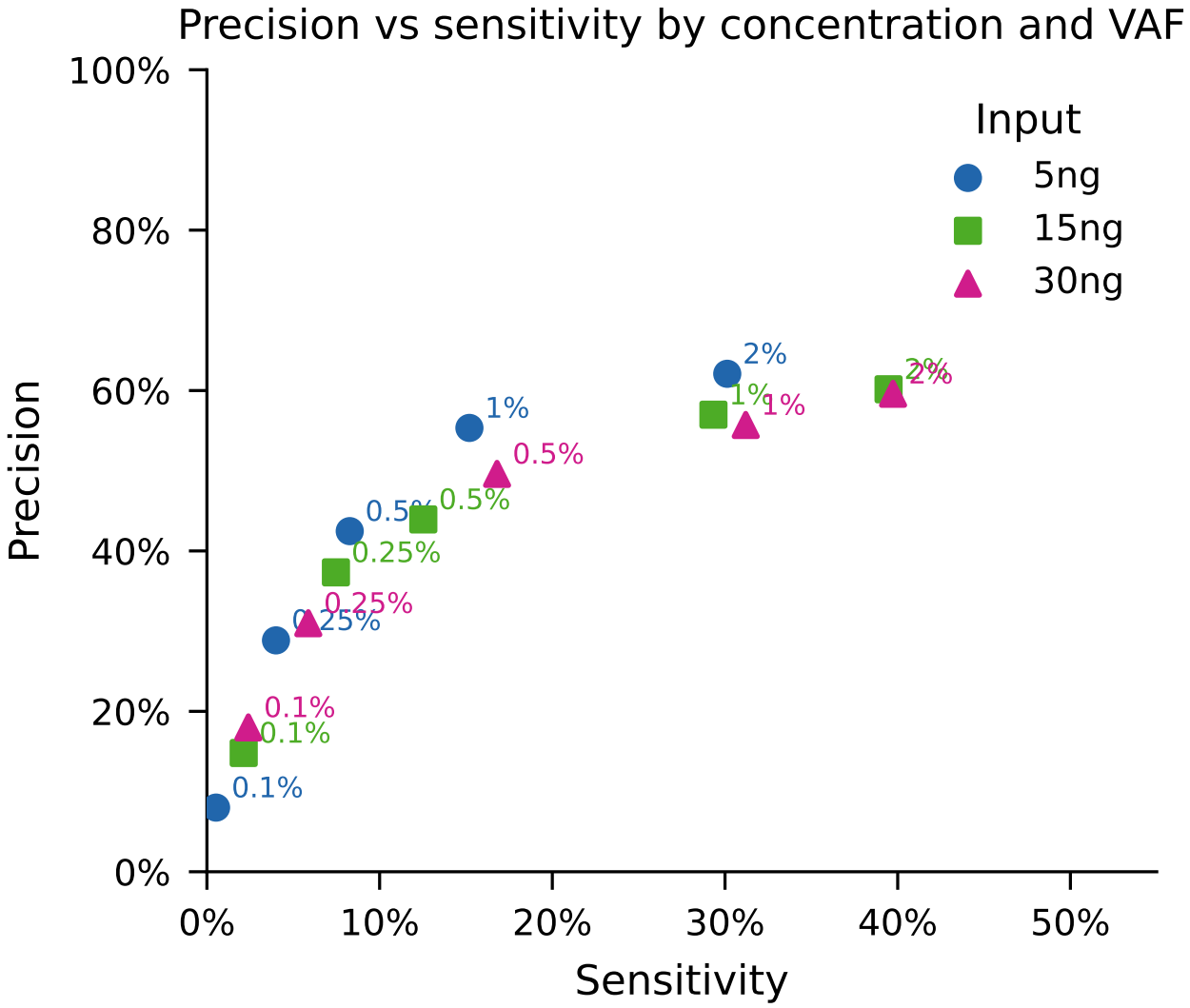

Figure 3: Precision vs sensitivity across all VAF/concentration combinations where at least one true positive was called. Each point is annotated with its VAF level. Precision is consistently above 0.37 wherever calls are made.